

Test Case	Conditions Identified	Initial Status	First Mismatch	Correction Made	Regression Tests Performed	Final Status
1	Duplicate machines, invalid temperature limits, unknown machine, duplicate event ID, out-of-order timestamp, invalid status, invalid defect count and anomalies.	Pass	Not Applicable	Not Applicable	Cases 2–9 checked again	Pass
2	Boundary temperature/vibration values, zero production and zero defects.	Pass	Not Applicable	Not Applicable	Cases 1, 3–9 checked again	Pass
3	Low-temperature and high-vibration anomalies; accepted events.	Pass	Not Applicable	Not Applicable	Cases 1, 2, 4–9 checked again	Pass
4	Out-of-order timestamp, unknown machine, invalid status and negative vibration.	Pass	Not Applicable	Not Applicable	Cases 1–3, 5–9 checked again	Pass
5	Multiple anomaly events, high vibration, out-of-order event, unknown machine and statistics.	Difference	Expected console shows 1 vibration anomaly event, but the written rule counts both accepted high-vibration events.	No code change. Kept the implementation according to the written rule.	Cases 1–4 and 6–9 checked again	Pass*
6	Multiple machines, boundary values, anomalies, unknown machine, invalid status, out-of-order timestamp and duplicate event ID.	Pass	Not Applicable	Not Applicable	Cases 1–5, 7–9 checked again	Pass
7	100 machines and 10,000 events; large-data processing and statistics.	Pass	Not Applicable	Not Applicable	Cases 1–6, 8–9 checked again	Pass
8	500 machines and 200,000 events; very large-data processing and statistics.	Pass	Not Applicable	Not Applicable	Cases 1–7 and 9 checked again	Pass
9	Duplicate machine ID, invalid machine, duplicate event ID, unknown machine, invalid status, negative vibration, invalid defect, invalid timestamp and incomplete/invalid events.	Difference	Expected output conflicts with the written rule that all records with a duplicate Machine ID must be rejected.	No code change. Kept the implementation according to the written rule.	Cases 1–8 checked again	Pass*